



**SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)**

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)  
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE  
Batch : 2025-29

### **Assignment :**

**Subject: Programming with Java**

**ASSIGNMENT No : 6**

**TITLE : Write a Java program to demonstrate the use of an inner class, an anonymous class.**

### **Problem Statement**

Write a Java program to demonstrate the use of an inner class, an anonymous class.

### **Learning Objectives**

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

### **Theory**

An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling.



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## Code and Output Screenshots

```
J Outer.java > ...
1  class Outer {
2      class Inner {
3          void display() {
4              System.out.println("Inner class method");
5          }
6      }
7
8      Run main | Debug main | Run | Debug
9      public static void main(String[] args) {
10         Outer.Inner obj = new Outer().new Inner();
11         obj.display();
12     }
13
14
```

PROBLEMS 22 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t...) Code

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Outer.java && java Outer
Inner class method

[Done] exited with code=0 in 0.484 seconds
```

## Conclusion

This program demonstrates the use of an inner class to organize related functionality within an outer class and an anonymous class to implement behavior without creating a separate named class. These features help make Java programs more modular, concise, and easier to maintain.



### Exercise

1. Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

Ans)

```
J Vehicle.java > Language Support for Java(TM) by Red Hat > Vehicle
1  interface Action {
2      void run();
3  }
4
5  public class Vehicle {
6
7      // Inner class
8      class Details {
9          void display() {
10             System.out.println("Vehicle: Car");
11         }
12     }
13
14     Run main | Debug main | Run | Debug
15     public static void main(String[] args) {
16
17         Vehicle v = new Vehicle();
18
19         // Inner class object
20         Details d = v.new Details();
21         d.display();
22
23         // Anonymous class
24         Action a = new Action() {
25             public void run() {
26                 System.out.println("Vehicle is moving");
27             }
28         };
29
30         a.run();
31     }
}
```

PROBLEMS 22 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeText, t...) Code

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Vehicle.java && java Vehicle
Vehicle: Car
Vehicle is moving

[Done] exited with code=0 in 0.487 seconds
```



2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

Ans)

```
FoodDelivery.java > Java > FoodDelivery
1  interface Delivery {
2      void status();
3  }
4
5  public class FoodDelivery {
6
7      // Inner class
8      class Order {
9          void details() {
10             System.out.println("Pasta Ordered");
11         }
12     }
13
14     Run main | Debug main | Run | Debug
15     public static void main(String[] args) {
16
17         FoodDelivery f = new FoodDelivery();
18
19         // Inner class
20         FoodDelivery.Order o = f.new Order();
21         o.details();
22
23         // Anonymous class
24         Delivery d = new Delivery() {
25             public void status() {
26                 System.out.println("Order Delivered");
27             }
28         };
29         d.status();
30     }
31 }
```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !excludeText, t...

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac FoodDelivery.java && java FoodDelivery
Pasta Ordered
Order Delivered

[Done] exited with code=0 in 0.537 seconds
```